

Pouya Mansournia

Senior Robotics Systems Engineer

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 Portfolio 📍 Türkiye | Open to Relocation



SUMMARY

Senior Robotics Systems Engineer with 8+ years of experience leading large-scale robotics, warehouse automation, and autonomous systems across logistics, industrial, and biomedical domains. Delivered production systems supporting 600K+ daily throughput and ultra-precision robotic platforms with international deployment.

Specialized in robotics architecture, embedded systems, and end-to-end autonomous system development from prototype to production. Co-founder-level ownership across the full product lifecycle, including hardware, software, and cloud integration.

Led cross-functional teams (15+) and \$500K–\$2M programs, achieving up to 50% cost reduction through in-house system design. Fluent in English, conversational Turkish. Open to relocation.

Professional Experience

Co-Founder & CINO | Product & Innovation Lead

X-Robotiics (IoT workplace intelligence startup, 20+ enterprise deployments) 

2024 – Present
Istanbul, Türkiye

- Architected and launched a production-grade IoT platform deployed across 20+ enterprise environments, defining end-to-end system topology from edge devices through cloud observability with zero critical production failures.
- Owned full technical roadmap: edge computing devices, distributed sensing infrastructure, MQTT communication, telemetry pipelines, and cloud monitoring systems.
- Engineered an offline-first resilience layer enabling uninterrupted edge-device operation and reliable cloud synchronization under unstable network conditions, a key differentiator for enterprise deployments.
- Directed end-to-end product lifecycle: system architecture, manufacturing coordination, calibration workflows, production validation, and enterprise client onboarding.
- Advised enterprise clients on warehouse automation strategy, AMR/AGV fleet sizing, deployment planning, and robotics infrastructure optimization.

Senior Manager, Robotics Infrastructure & Automation

Digikala (Iran's largest e-commerce platform, comparable to Amazon — 600K+ daily parcels) 

2021 – 2024
Tehran, Iran

- Owned architecture and deployment of robotics automation infrastructure across 3 large-scale fulfillment centers, supporting 600K+ daily parcel throughput in 24/7 production environments. Led a cross-functional engineering team of 15+ and managed \$500K–\$2M hardware/software programs.
- Scaled modular robotic sortation systems from ~2,000 to 6,000+ parcels/hour using distributed PLC control, embedded real-time telemetry pipelines, and sensor fusion, achieving zero unplanned downtime over 18 months.
- Delivered autonomous AMR/AGV platforms integrating SLAM-based navigation, sensor fusion, fleet orchestration, and safety-critical motion control across multi-kilometer fulfillment facilities.
- Improved operational throughput by ~20% and reduced fulfillment errors from ~5% to near-zero via human-in-the-loop automation and closed-loop quality monitoring systems.
- Reduced capital expenditure by ~50% (~\$1M+ savings) through modular hardware standardization and in-house actuator development, enabling 3x deployment scale within the same budget.
- Led cross-functional initiatives spanning robotics, embedded systems, backend infrastructure (Python, Linux, Git), mechanical systems, and operations, including technical direction and hiring.

Principal Mechatronics Systems Engineer

TARFAND Technical Solutions (biomedical robotics startup, exported to Turkey & Russia) [↗](#)

2019 – 2021
Tehran, Iran

- Led development and international commercialization of Iran's first domestically produced IVF micromanipulation robotic platform, a system with no domestic precedent and 60% lower cost than comparable European platforms.
- Designed and validated sub-micron motion control systems, achieving 0.2 μm positioning accuracy, a 10x improvement over incumbent platforms using piezoelectric actuation and deterministic RTOS-based control loops.
- Architected closed-loop motion-control systems integrating tremor suppression, strain-gauge feedback, and deterministic embedded control (STM32, CAN Bus, custom PCB design with high-voltage piezo interfaces).
- Commercialized and deployed 38+ biomedical robotic systems across domestic and international markets, validating hardware-software co-design methodology in a regulated clinical environment.

Senior Mechanical Systems Engineer | Industrial Automation & Mobile Systems

Barak.co (Large-Scale Bakery Production Systems Manufacturer) [↗](#)

2018 – 2019
Tehran, Iran

- Designed and deployed industrial automation systems across 500+ large-scale food-processing units using synchronized multi-stage mechanical transmission architectures.
- Led engineering and field deployment of mobile disaster-response kitchen platforms across 60+ emergency-response units, including safety-critical towing, stabilization, and fail-safe emergency braking systems.
- Conducted FMEA-driven reliability engineering and failure analysis across structural, thermal, and mechanical subsystems.

Robotics Systems Architect & Research Assistant

Syntech Technology and Innovation Center [↗](#)

2015 – 2018
Tehran, Iran

- Led robotics architecture and technical mentorship for multidisciplinary robotics teams across RoboCup and applied robotics projects.
- Architected autonomous mobile robotic platforms integrating omnidirectional mobility, robotic manipulation, embedded electronics, and ROS-based autonomy systems.
- Developed STM32-based embedded control and power electronics systems featuring CAN communication, fault protection, and distributed robotics control architectures.
- Engineered robotic manipulator systems with closed-loop motion control, kinematic optimization, and sub-5 mm positioning accuracy.
- Guided cross-functional robotics teams through iterative design reviews, improving engineering quality, manufacturability, and system reliability.

Technical Skills

Robotics & Autonomy: ROS/ROS2 | SLAM | Autonomous Navigation | Fleet Management | Motion Control | Sensor Fusion | AGV/AMR Systems | Robotics Middleware | Gazebo

Mechanical & Simulation: SolidWorks | Abaqus | ANSYS | MSC Adams | GD&T | DFM/DFA

Software & Infrastructure: Python | C/C++ | Linux | Git | MQTT | Telemetry Systems | Distributed Systems | Edge Computing

Embedded & Controls: STM32 | Embedded Linux | RTOS | CAN Bus | PID Control | Motor Drivers | Real-Time Systems | PCB Design | Piezoelectric Actuation

Leadership: Cross-functional team leadership (15+ engineers) | Program budget ownership (\$500K–\$2M) | Technical roadmap | Hiring | OKR/KPI definition | System architecture | Vendor management

Education

M.Sc. Mechanical Engineering - Applied Design

Qazvin Islamic Azad University (QIAU)

2019 – 2021
Qazvin, Iran

- GPA: 3.73/4 | Ranked #1 among all Mechanical Engineering graduate students
- **Thesis:** Design and analysis of a nano-precision fast steering mirror platform using flexure mechanisms and piezoelectric actuation.

B.Sc. Mechanical Engineering - Applied Design

Qazvin Islamic Azad University (QIAU)

2014 – 2018
Qazvin, Iran

- Thesis: Design and application of strain wave gearing systems for 6-DoF robotic manipulators.

Languages

Persian

Native

English

TOEFL iBT: 95

Turkish

Conversational

Honors & Recognition

- Ranked 1 among graduate students of Mechanical Engineering, QIAU (2021)
- Member of Iran's National Elites Foundation (INEF)
- First Place – RoboCup @Work League World Competition, **Sydney** (2019)
- First Place – RoboCup Asia Pacific Competition (2018)
- First Place – IranOpen RoboCup Competition (2018)
- Third Place – RoboCup World Competition, **Montreal, Canada** (2018)
- First Place – RoboCup Asia Pacific Competition, **Thailand** (2017)
- First Place – Rescue Robot League World Competition, **China** (2015)